

Problem 1 — Implement a Perceptron from Scratch

Dataset: AND / OR gate

Create a perceptron **without TensorFlow/PyTorch/Keras/sklearn**.

Use:

```
X = [  
    [0,0],  
    [0,1],  
    [1,0],  
    [1,1]  
]
```

Train it for:

1. AND gate
2. OR gate

You have to implement:

- weights initialization
- bias
- weighted sum
- step activation
- prediction
- error calculation
- weight update
- bias update
- training loop

Then print:

Epoch 1:

Weights = ...

Bias = ...

Errors = ...

Epoch 2:

...

Problem 2 — Diabetes Prediction using NN

Dataset: Diabetes Dataset

Tasks to be done after loading the data

Data Understanding & Preprocessing

- Display the shape, data types and summary statistics.
- Check for missing values.
- Identify whether any columns contain invalid zero values.
- Separate input features and target variables.
- Check the class distribution of the target.
- Split the data into training and testing sets.
- Identify whether numerical features have significantly different scales.
- Apply appropriate feature scaling.

Neural Network

- Build a binary classification neural network using Keras.
- Use at least one hidden layer.
- Use an appropriate activation function for the hidden layer.
- Use an appropriate activation function for the output layer.
- Use Binary Cross Entropy as the loss function.
- Train the model and plot training vs validation loss.
- Plot training vs validation accuracy.

Gradient Descent Experiment

- Train the same network using:
 - SGD
 - Mini-Batch Gradient Descent and Batch Gradient Descent.

Regularization

- Add Dropout to the network.
- Train models with:
 - Dropout = 0
 - Dropout = 0.2
 - Dropout = 0.5
- Compare training and validation performance.
- Determine whether the original model was overfitting.

Early Stopping

- Train the model for a sufficiently large number of epochs.
- Implement Early Stopping using validation loss.
- Experiment with different patience values.
- Identify the epoch at which the best model was obtained.
- Compare the model with and without Early Stopping.

Problem 3 — Student Performance Prediction

Dataset : Student Performance

Tasks

A. Preprocessing

- Load the dataset.
- Check missing values.
- Identify numerical and categorical variables.
- Identify the target variable.
- Encode categorical variables.
- Apply feature scaling.

B. Regression Neural Network

- Split the dataset into train, validation and test sets.
- Build a neural network for regression.
- Experiment with different numbers of hidden layers.
- Experiment with different numbers of neurons.
- Use an appropriate regression loss function.
- Plot training and validation loss.

C. Gradient Descent

- Train the model using different optimizers.
- Compare SGD, Mini-Batch SGD and Batch.

D. Overfitting

- Create a relatively large network.
- Train it without regularization.
- Identify evidence of overfitting.
- Add Dropout.
- Compare performance before and after Dropout.

E. Early Stopping

- Train for a large number of epochs.
- Apply Early Stopping.
- Restore the best weights.
- Compare:
 - Without Early Stopping
 - Vs
 - With Early Stopping